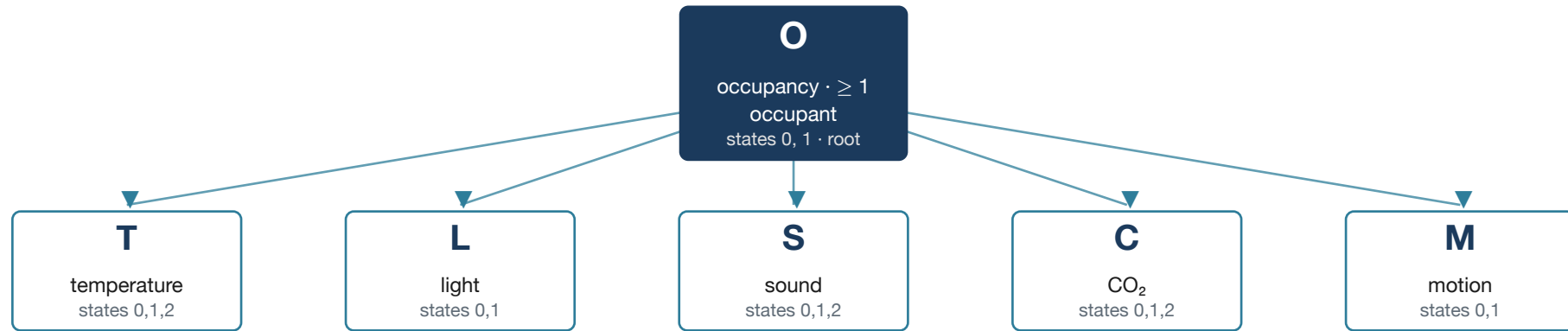


Model and assumption

A naive-Bayes network over one occupancy state and five discretised sensors

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FACTORISATION — 5 EDGES, 6 NODES

$$P(O, T, L, S, C, M) = P(O) \prod_{X \in \{T, L, S, C, M\}} P(X | O)$$

Occupancy is the single common cause; each sensor is a leaf with one parent. Every arrow points **from** the hidden state **to** the observed reading.

THE MODELLING ASSUMPTION WE ACCEPT

The five sensors are **conditionally independent given O**: e.g. $L \perp\!\!\!\perp C \mid O$.

This is an approximation, not a measured fact. Light and CO₂ also share time-of-day drivers, so residual correlation is absorbed as if it were independent evidence — a point we return to on slide 3.

Protocol and evidence

One fitted model, one test set, three evidence conditions

FIXED CHRONOLOGICAL SPLIT

Train 7 090 rows (22→26 Dec 2017)

O=0: 5 483 · O=1: 1 607

Test 3 039 rows (26 Dec→11 Jan)

O=0: 2 745 · O=1: 294

No shuffling. Bins fitted on the training period only; CPDs by Dirichlet posterior mean, one pseudocount per cell.

DECISION RULE, HELD FIXED

Declare occupied when $P(O = 1 | e) \geq 1/6$.

Costs: FP = 1, FN = 5, correct = 0.

$$\text{mean loss} = (\text{FP} + 5 \cdot \text{FN})/n$$

Fitted prior $P(O = 1) = 0.2267 \geq 1/6$, so the prior-only baseline declares **every** record occupied.

ONE NUMBER, READ CAREFULLY

$$P(M = 1 | O = 1) = \frac{939+1}{1607+2} = 0.5842$$

Numerator: 939 training rows with motion, +1 Dirichlet pseudocount. **Denominator:** 1 607 rows in the parent configuration O=1, +1 pseudocount for **each** of M's two states. Hand count and pgmpy agree to 1e-12.

SAME MODEL, SAME 3 039 TEST RECORDS — ONLY THE EVIDENCE CHANGES

Condition	Brier	Mean loss	Acc.	Prec.	Recall	TN	FP	FN	TP
Prior only (no sensors)	0.10428	0.90326	0.0967	0.0967	1.0000	0	2 745	0	294
Full sensors	0.02350	0.08292	0.9829	0.9919	0.8299	2 743	2	50	244
Without L (our outage)	0.01118	0.04673	0.9546	0.6814	0.9966	2 608	137	1	293

The **full sensors** → **without L** pair isolates the outage: identical CPDs, records, threshold and costs. Losing L trades **+135 false positives** for **−49 false negatives** at a 5:1 cost ratio that **lowers** mean loss from 0.08292 to 0.04673, even though accuracy and precision fall.

Engineering conclusion

What the outage costs us, what this evidence cannot settle, and what to run next

OUTAGE CONSEQUENCE — ASSIGNED INCIDENT

Record **8929**, 2018-01-10 22:41

full: {T=0, L=0, S=1, C=0, M=0} → without L: {T=0, S=1, C=0, M=0}

	$P(O = 1 e)$	decision
full sensors	5.07×10^{-9}	empty
without L	3.51×10^{-6}	empty

The posterior rises by a factor of about 700, yet stays far below 1/6: **this** decision is unchanged. The outage degrades confidence, not the verdict — the four surviving sensors already agree. Aggregate effect: recall 0.83 → **0.997**, precision 0.99 → **0.68**.

THE LIMITATION THAT MATTERS MOST

A lower loss without L is **not** evidence that the light sensor is harmful.

All 50 full-sensor false negatives occur at L=0 and none at L=1. Under the naive-Bayes assumption, L=0 multiplies in as **independent** evidence for “empty”, overwhelming the other sensors and pushing genuinely occupied night-time records below the threshold. Dropping L removes an over-counted factor — it does not remove a bad sensor.

The result is tied to this split, this discretisation, this 1/6 threshold and this 5:1 cost ratio. It is not causal, and it does not transfer to another building or season.

NEXT VALIDATION

Rolling-origin evaluation over successive day boundaries, with every modelling choice frozen — if “without L” keeps winning across folds **and** across a sweep of cost ratios, the finding is about the independence assumption, not about this one 17-day test window. A dependency-aware structure (an edge $L \rightarrow C$, or a time-of-day parent) is the direct test of the same hypothesis.